

RTutor.ai Quick-Start Guide

Grouped Summaries, Grouped Density Curves, Ratio SD, Effect Size, and Line Graphs
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0) Setup

Use the same Exam Grades dataset from Guide 1. If you are starting fresh:

1. Upload the dataset in RTutor.ai under **(1. Data) -> Browse**.
2. Confirm four columns: *Student ID*, *Year In School*, *Practice Exam Grade*, *Actual Exam Grade*.
3. Under **(2. Data Types) -> Edit**, set Student ID = Character, Year In School = Category, both grade columns = Integer (or Numeric).

1) Grouped Summary Table

Goal: Generate summary statistics for Practice Exam Grade grouped by Year In School.

Your prompt to RTutor.ai must:

1. Reference the Exam Grades dataset, the variable Practice Exam Grade, and the grouping variable Year In School.
2. Request these statistics for each group: count, mean, standard deviation, min, Q1, median, Q3, max.
3. Ask for rounding to two decimal places, a consistent display order, and a clear table layout with one row per group.

Check: Does the output show separate rows for Freshmen and Upperclassmen, each with all eight statistics?

2) Grouped Density Curves

Goal: Create overlapping density curves of Practice Exam Grade, one for each Year In School group, on a single plot.

Your prompt must include:

1. Specify the Exam Grades dataset, the variable Practice Exam Grade, and group by Year In School.
2. Say “grouped density plot” or “overlapping density curves by Year In School” (add “not a histogram” if needed).
3. Set x-axis limits 0 to 100, tick marks every 5 points, font size 14, and a color palette of your choice. Place the legend at the bottom.
4. Ask for different colors and/or fill transparency so both curves are visible where they overlap.

Check: Are there two distinct density curves on the same plot? Can you tell which group is which from the legend? Are the overlapping areas visible (not hidden)?

3) Ratio SD and Effect Size

Goal: Compute the ratio of standard deviations and the effect size for Practice Exam Grade between the two Year In School groups.

Your prompt must include:

1. **Ratio SD:** Ask RTutor.ai to compute the standard deviation of Practice Exam Grade for each Year In School group, then divide the larger SD by the smaller SD. Ask it to label the result as “Ratio SD.” A ratio near 1 means the groups have similar spread; a ratio above 2 suggests different variability.

2. **Effect Size (Difference in Means):** Ask RTutor.ai to compute the Cohen's d Effect Size for the difference in average Practice Exam Grade between Freshman and Upperclassman (e.g., Upperclassman mean – Freshman mean). Ask it to label the result as “Effect Size” and round to two decimals. This tells you the practical size of the gap between the two groups in standard deviations.

Check: Did RTutor.ai report both the Ratio SD and the Effect Size with clear labels? If it struggles, break these into two separate prompts.

4) Line Graph

Goal: Create a line graph comparing group means across both exam types (Practice vs Actual) for Freshman and Upperclassman.

Your prompt must include:

1. Ask RTutor.ai to compute the mean of Practice Exam Grade and the mean of Actual Exam Grade for each Year In School group.
2. Ask for a line graph with x-axis = Exam Type (Practice, Actual), y-axis = Mean Grade, and separate lines (with markers) for Freshman and Upperclassman.
3. Ask for “error bars” indicating the 95% margin of error.
4. Request font size 14, a color palette of your choice, and the legend at the bottom.

Check: Are there two lines (one per group) connecting two points each? Does each point have bars extending above and below the point (indicating variability and uncertainty in estimation)? Can you see which group improved more from Practice to Actual?

5) Interpretation & Extension

1. Write 2–3 sentences: Compare the two groups' Practice Exam Grade distributions (center, spread). State the Ratio SD and the effect size, and explain what they imply about any group differences.
2. Using the line graph, describe whether both groups improved from Practice to Actual and whether one group improved more than the other.
3. Repeat the full workflow (grouped summary → grouped density → Ratio SD and effect size → line graph) using Actual Exam Grade as the primary variable, and write the same interpretation sentences.

6) Troubleshooting

1. **Wrong variable / not found:** Use exact column names (Student ID, Year In School, Practice Exam Grade, Actual Exam Grade).
2. **Only one group shown:** Make sure Year In School is set to Category in (2. Data Types). Explicitly say “group by Year In School.”
3. **Wrong graph type:** Say “grouped density plot, not a histogram” or “line graph, not a bar graph.”
4. **Prompt partially ignored:** Break long prompts into smaller steps: grouped table -> density curves -> Ratio SD -> effect size -> line graph.

Still stuck? Save your prompt and a screenshot, then ask a CA or instructor.